

## Image Analysis Report

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| Source    | Valve-Vorrosion.jpg                  |
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### Executive Summary

The valve assembly exhibits advanced, high-severity atmospheric corrosion characterized by significant oxide scale formation and material loss. The degradation is severe enough to likely compromise the mechanical integrity of the handwheel and fasteners, posing a risk of operational failure.

### Key Findings

| # | Finding                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | Extensive atmospheric corrosion (rust) covering 100% of the visible valve and handwheel surfaces.                           |
| 2 | Significant scaling and flaking of corrosion products, indicating deep penetration and metal loss.                          |
| 3 | Bonnet bolts and nuts are heavily encrusted with corrosion, likely seizing the threads and preventing standard maintenance. |
| 4 | The valve handwheel shows signs of structural thinning, which may lead to failure if high torque is applied.                |
| 5 | Surface corrosion is visible on the underlying pipe, though depth of penetration cannot be determined visually.             |
| 6 | Heavy accumulation of corrosion products around the valve stem entry suggests the packing and stem are likely seized.       |

## Localized detections (VLM, approximate)

- Corroded Handwheel (conf. 96%): x=0.280, y=0.220, w=0.200, h=0.480
- Severe Bonnet Corrosion (conf. 94%): x=0.440, y=0.300, w=0.350, h=0.320
- Pipe Surface Corrosion (conf. 89%): x=0.000, y=0.840, w=1.000, h=0.150

## Recommendations

| # | Recommendation                                                                                                                     |
|---|------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Isolate the process line and schedule the replacement of the entire valve unit due to the severity of the corrosion.               |
| 2 | Perform Ultrasonic Thickness (UT) testing on the adjacent piping to ensure the remaining wall thickness meets safety standards.    |
| 3 | Do not attempt to operate the handwheel until the mechanical integrity is verified to prevent sudden fracture.                     |
| 4 | Utilize penetrating oils or induction heating for fastener removal if disassembly is required for further forensic analysis.       |
| 5 | Apply a high-performance marine-grade coating system (e.g., zinc-rich primer and epoxy topcoat) to all new components.             |
| 6 | Review the local environment for corrosive agents (e.g., salt spray or chemical fumes) that may have accelerated this degradation. |
| 7 | Establish a more frequent inspection interval for this specific location within the corrosion management program.                  |